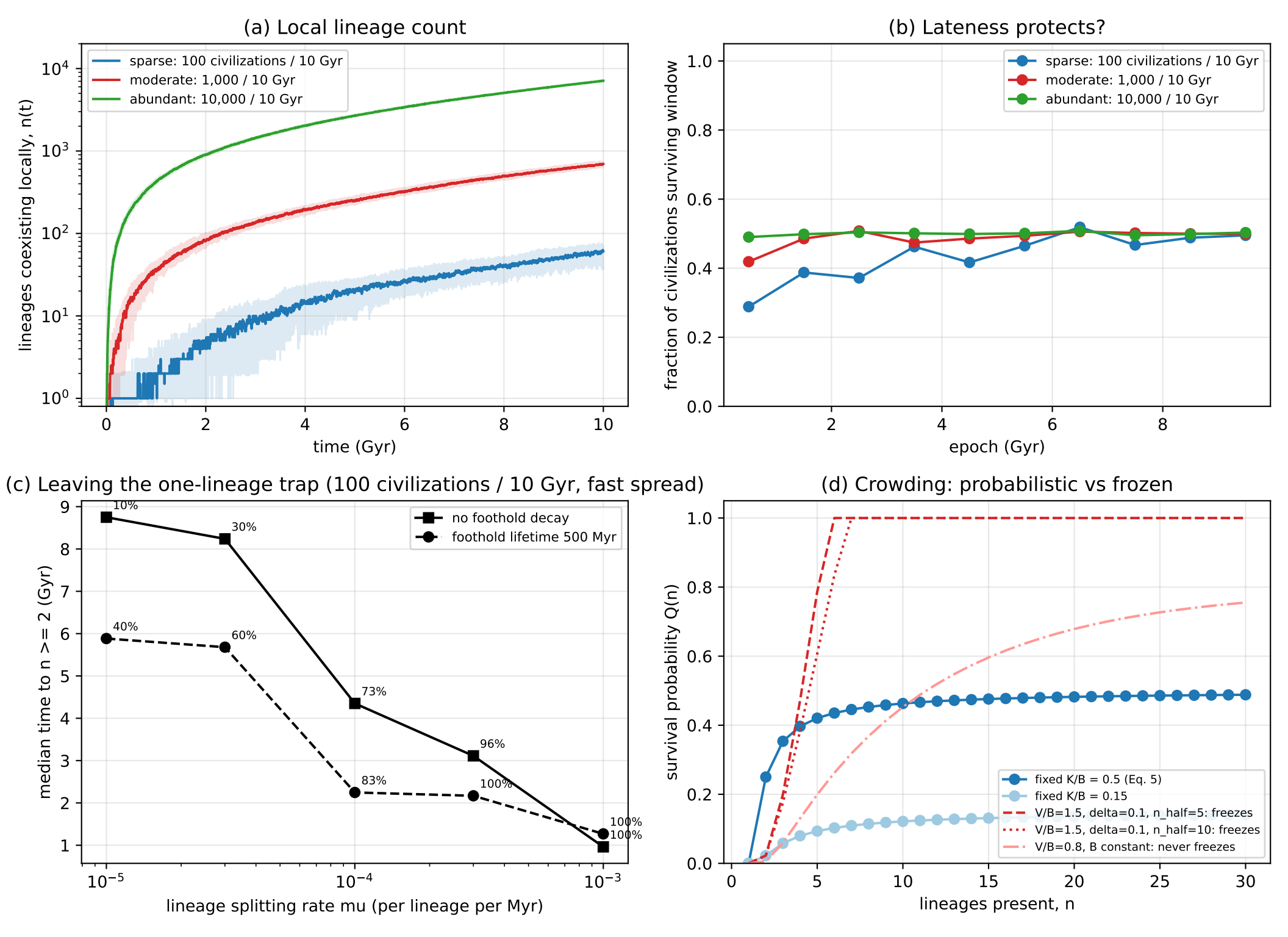

(a) Local lineage count
sparse: 100 civilizations / 10 Gyr
moderate: 1,000 / 10 Gyr
abundant: 10,000 / 10 Gyr
lineages coexisting locally, n(t)
time (Gyr)
(b) Lateness protects?
fraction of civilizations surviving window
epoch (Gyr)
(c) Leaving the one-lineage trap (100 civilizations / 10 Gyr, fast spread)
no foothold decay
foothold lifetime 500 Myr
median time to n >= 2 (Gyr)
lineage splitting rate mu (per lineage per Myr)
10%
30%
73%
96%
100%
40%
60%
83%
100%
(d) Crowding: probabilistic vs frozen
survival probability Q(n)
lineages present, n
fixed K/B = 0.5 (Eq. 5)
fixed K/B = 0.15
V/B=1.5, delta=0.1, n_half=5: freezes
V/B=1.5, delta=0.1, n_half=10: freezes
V/B=0.8, B constant: never freezes